

NUSSINOV ALGORITHM FOR RNA SECONDARY STRUCTURE PREDICTION

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4. NUSSINOV ALGORITHM (OR BASE PAIR MAXIMIZATION)

- Foundation of RNA structure prediction
- Attempts to maximize RNA base pairs
- Similar to DP algorithms we have already studied
- Does not predict pseudoknots



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BASIC IDEA

- Given an RNA sequence of some length
 - If there are two complimentary bases, score is 1
 - Else, score is 0*
- We find the maximum number of base pairs that are possible in a given sequence



Alignment Method

- Alignment Method
 - Align RNA strand to itself
 - Score increases for feasible base pairs
 - ~~Do not~~ consider G/U for this example
We will
- Each score independent of overall structure
- Bifurcation adds extra dimension (don't worry about this)

G - C

A - U

G - U wobble



	A_1	A_2	U_3	C_4	U_5	G_6	U_7	U_8	A_9	C_{10}	G_{11}	C_{12}	A_{13}
A_1													
A_2													
U_3													
C_4													
U_5													
G_6													
U_7													
U_8													
A_9													
C_{10}													
G_{11}													
C_{12}													
A_{13}													



	5'	A ₁	A ₂	U ₃	C ₄	U ₅	G ₆	U ₇	U ₈	A ₉	C ₁₀	G ₁₁	C ₁₂	A ₁₃ 3'
→	A ₁	0	0	1	1	2	2	2	3	3	3	3	3	4
→	A ₂	0	0	1	1	1	1	2	2	2	2	3	3	4
-	U ₃	0	0	0	0	0	1	1	1	2	2	3	3	4
	C ₄	0	0	0	0	0	1	1	1	2	2	3	3	4
	U ₅	0	0	0	0	0	1	1	1	2	2	3	3	4
	G ₆	0	0	0	0	0	0	1	1	1	2	2	3	3
	U ₇	0	0	0	0	0	0	0	0	1	1	2	2	2
	U ₈	0	0	0	0	0	0	0	0	1	1	1	1	2
	A ₉	0	0	0	0	0	0	0	0	0	0	1	1	1
	C ₁₀	0	0	0	0	0	0	0	0	0	0	1	1	1
	G ₁₁	0	0	0	0	0	0	0	0	0	0	0	1	1
	C ₁₂	0	0	0	0	0	0	0	0	0	0	0	0	0
	A ₁₃	0	0	0	0	0	0	0	0	0	0	0	0	0

U₇ — 4 — A₉
 G₆ — 3 — C₁₀
 U₅ — 2 — G₁₁
 C₄ — C₁₂
 U₃ — 1 — A₁₃
 A₁ — A₂
 5' — 3'

POINTS TO CONSIDER

- RNA structure prediction remains a difficult problem to solve
- We discussed 4 different approaches (there are many others) to solve RNA secondary structure elements,
- Which approach do you think is the better of the lot?
 1. Dot plot (visual)
 2. Minimum free energy (closer to reality?)
 3. Co-variance (bring evolution into the equation)
 4. Base pair maximization (considers all possibilities)



PRACTICE EXERCISE



	G	G	G	A	A	A	U	C	C
G									
G									
G									
A									
A									
A									
U									
C									
C									

